

Ritika Ratnu

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EDUCATION

- **KTH Royal Institute of Technology** Stockholm, Sweden
PhD - School of Electrical Engineering and Computer Sciences Jan 2022 - present
- **IISc Bangalore** Bengaluru, India
Master of Technology - Computational and Data Sciences August 2017 - May 2019
- **MNIT Jaipur** Jaipur, India
Bachelor of Technology - Electronics and Communication August 2012 - June 2016

SKILLS SUMMARY

- **Languages:** C++, Python, System Verilog, CUDA, x86 Assembly
- **Tools:** Synopsys and Cadence logic synthesis and physical implementation tools, Modelsim and Verdi(RTL logic debugging tools), VIM, TCSH shell

EMPLOYMENT

- **Intel India Pvt. Ltd.** Bengaluru
Pre-Silicon Architecture Engineer June 2019 - December 2022
 - **Modeling:** Modeling of Performance Monitors (PMONs) for next-gen Intel mid-level cache in the reference model.
 - **Validation:** Performance validation with expertise in Mid-Level cache and Memory Execution Unit.

RELEVANT PROJECTS

- **Composable Post-Layout Accurate Energy Models at Higher Abstractions (Ph.D 1st year)**
 - Building post-layout accurate energy-model that can be used to explore the design space at higher abstractions for mapping algorithms on hardware.
 - Learned how to implement a design from RTL-to-GDSII using Synopsys and Cadence logic and physical synthesis tools, debugging issues in design, and analyzing power results.
 - Developed infrastructure that makes the post-layout energy numbers available to High Level Synthesis tools.
 - One conference and one journal paper are expected from the work.
- **Performance Validation for multiple Next-Gen Intel Cores (Intel)**
 - RTL model is compared with cycle-accurate reference simulator for correlation of industry-standard benchmarks. Outliers with significant miscorrelation are debugged to find bugs or modeling differences.
 - As part of Big-Core PV team, point-of-contact for debugging performance differences in MLC and MEU cluster. Identified several critical performance bugs and suggested fixes for multiple gen Intel processors including AlderLake and MeteorLake.
 - Wrote code for workarounds and modeling gaps in the cycle-accurate simulator.
- **Modelling of Performance Monitoring Counters (Intel)**
 - Performance Monitoring counters are implemented inside the chip hardware and visible to the programmers to track core's activities.
 - Implemented these counters for MLC cluster in the cycle-accurate simulator to check viability in RTL.
- **An Accelerated Algorithm for Accurate De-Novo Genome Assembly (M.Tech Thesis Project)**
 - A parallel scalable algorithm to render accurate genome assembly by preserving mutations.
 - Algorithm reported < 1% misassembly, significantly lower than SPades (4.5%) and MEGAHIT (2.5%) compared on QUAST tool.

PUBLICATIONS

- Ritika Ratnu and Karthik Varadarajan, "New Methodology for Performance Validation of ML-based Hardware Prefetcher", in *Intel Design and Test Technology Conference (DTTC)*, 2021

MENTORING AND SUPERVISION

- Supervising two Master thesis students, Tong Lou, Xuying Chen at KTH (ongoing).
- Supervising a research work with Ruimin Shi, Masters student, KTH (ongoing).
- Mentor for Rahul Meena and Amit Bhakta at Intel, 2020-2021.